

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-6	Team No-20	Project Title-Rental & Sharing Economy Product Portal
Team Lead Roll Number		2520030030
Team Lead Name		UPPUTURI ROHIT SAI
Team Member Roll Number		2520030315
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1. Abstract

The “Rental & Sharing Economy Product Portal” is an online platform developed to enable users to rent products such as electronics, furniture, and tools for a specific duration. The growing demand for cost-effective and sustainable usage of products has led to the rise of rental-based systems instead of ownership.

This project provides a structured solution where users can browse a rental catalogue, select products, specify rental duration, and calculate deposits dynamically. It also includes features such as delivery details, condition reporting of products before and after use, and a return workflow to ensure accountability.

The system will maintain rental history, send notifications, and provide an admin panel for managing products and users. The application will be developed using modern frontend and backend technologies like React.js, Node.js, and REST APIs to ensure scalability and performance. This project aims to promote efficient resource utilization and provide a seamless user experience in the online rental ecosystem.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Rental Catalogue Duration Picker Deposit Calculation Delivery Form Condition Report	2520030030	
Return Workflow Rental History Notifications Admin Panel API Testing (Postman)	2520030315	

3. Module Details

- The **User Authentication Module** is used for handling user registration and login functionality. It is implemented using HTML, CSS, JavaScript for frontend validation and Node.js for backend processing.
- The **Product Listing Module** is responsible for displaying all available rental products to users. It is developed using React.js for dynamic UI and Node.js APIs to fetch product data.
- The **Product Details Module** provides detailed information about each product, including price, description, and rental options. This module uses React.js and JavaScript for interactive display.
- The **Rental Booking Module** allows users to select rental duration and book products. It includes cost and deposit calculation using JavaScript and backend handling with Node.js.
- The **Payment Module** manages the payment process and deposit handling. It is implemented using basic JavaScript logic and Node.js for processing transactions.
- The **Delivery & Return Module** handles the delivery tracking and return workflow of rented products. It uses React.js for frontend updates and Node.js for managing status.
- The **Admin Module** enables the admin to manage products, users, and bookings. It is built using React.js for the dashboard and Node.js for backend operations.

4. Frameworks / Platforms / Software Used

Frontend:

- * HTML5, CSS3, JavaScript (ES6+)
- * React.js (Hooks, React Router)
- * Tailwind CSS / Bootstrap

Backend:

- * Node.js
- * Express.js

Database:

- * MySQL (optional) / LocalStorage

API & Communication:

- * REST APIs
- * Axios

Testing:

- * Postman

Guide Name and Signature (remarks if any) _____